# TNA in Neuroscience: Throughput, Incoherence, and the Structural Basis of Neural Stability

**Claudio Bresciano** Independent Researcher Theorem of Axiomatic Necessity (TNA)

## "Disclaimer / Warning

**This article is intended for illustrative and scientific purposes only, demonstrating how TNA variables (G, Ψ, ℵ) have real neural correlates. It is not medical advice and does not constitute treatment or diagnosis for any neurological or psychiatric condition.**

Always consult a licensed physician, neurologist, psychiatrist, or qualified healthcare professional before making changes to your lifestyle, therapy, or treatment. Individual brain responses vary; TNA is a conceptual framework, and personal safety must take priority."

## Abstract

The human brain is traditionally modeled as an information-processing device governed by synaptic plasticity, network dynamics, and neurochemical modulation. This paper proposes an alternative framework based on the Theorem of Axiomatic Necessity (TNA), viewing the brain as an open non-equilibrium system where neural coherence requires continuous throughput (Ψ) to discharge incoming incoherence (G). When G exceeds Ψ, accumulated incoherence (ℵ) leads to functional collapse, manifesting as neurodegenerative diseases, psychiatric disorders, or cognitive fatigue. Empirical phenomena—from Alzheimer's plaque accumulation to depression-related rumination—are reframed as throughput saturation rather than primary deficits. The model is falsable, predictive, and compatible with existing neuroscientific data, offering a unified explanation for neural persistence and failure.

Keywords: Theorem of Axiomatic Necessity (TNA); neural throughput; incoherence accumulation; neurodegeneration; psychiatric disorders; non-equilibrium brain dynamics

## 1. Introduction: The Brain as a Household Plumbing System

Consider a common household shower supplied by a rooftop tank and municipal pressure. Water enters (G), accumulates in the tank (ℵ), and discharges through the showerhead (Ψ). If input exceeds discharge, the tank overflows or pressure drops; if discharge is blocked, the system fails catastrophically.

The human brain operates under the same structural constraint:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

where G is sensory/cognitive input, Ψ is neural processing/discharge (firing, pruning, clearance), and ℵ is accumulated noise (misfolded proteins, chronic inflammation, unresolved loops).

This paper applies TNA to neuroscience, showing that many "disorders" are not primary pathologies but consequences of throughput imbalance.

## 2. Core TNA Mapping to Neural Systems

| TNA VARIABLE | NEURAL EQUIVALENT | BIOLOGICAL MANIFESTATION |
|---|---|---|
| G (generation) | Sensory input, stress, neurochemical load | Cortisol spikes, excitatory glutamatergic drive |
| Ψ (throughput) | Synaptic processing, glymphatic clearance, pruning | Sleep-dependent amyloid removal, BDNF-mediated plasticity |
| ℵ (incoherence) | Protein aggregates, chronic inflammation, unresolved loops | Amyloid plaques, rumination circuits, glial activation |
| F_min (minimal coherence) | Baseline homeostasis required for viability | Blood-brain barrier integrity, mitochondrial efficiency |

## 3. Specific Applications

3.1 **Neurodegeneration (Alzheimer's, Parkinson's)** G high (aging-related protein production) > Ψ low (glymphatic slowdown in old age). ℵ accumulates as plaques/tangles → cascade collapse. Application: Deep sleep or exercise increases Ψ (clearance) —reduces risk 20-40% (empirical studies).

3.2 **Depression and Anxiety** G high (rumination, chronic stress). Ψ low (prefrontal-hippocampal saturation). ℵ as "stuck" negative loops. Application: Exercise/meditation increases Ψ (BDNF, neurogenesis) —symptom reduction 40-60%.

3.3 **Cognitive Fatigue and Burnout** G high (information overload). Ψ saturated (no recovery). ℵ as mental fog. Application: Rest intervals increase Ψ —prevents collapse (ultradian rhythm studies).

3.4 **Sleep and Memory Consolidation** During sleep, G low, Ψ high (slow-wave clearance). ℵ discharged → coherence restored. Application: Sleep deprivation = ℵ accumulation → memory impairment, Alzheimer risk.

3.5 **Neuroplasticity and Learning** High Ψ (childhood pruning) = low ℵ → rapid learning. Adult Ψ lower → learning slower. Application: Exercise/sleep boosts Ψ —restores plasticity (adult neurogenesis data).

## 4. Predictive Power and Falsability

- Predicts: Interventions that increase Ψ (sleep, exercise, ketosis) delay neurodegeneration.
- Falsable: If Ψ enhancement fails to reduce ℵ in controlled trials, model weakens.
- Distinguishes from traditional models: Not "broken circuits" —throughput mismatch.

## 5. Figure: Neural Sacrifice Rate

[Descripción para generar con SimpAI/Flux: "Curva de sacrifice rate neuronal: eje X sacrifice rate (low to high), eje Y coherence (optimal band in middle), zonas left 'accumulation collapse' (Alzheimer), right 'over-pruning collapse' (autism-like). Estilo científico limpio."]

Interpretation: Optimal coherence at intermediate sacrifice (pruning/clearance) —too low = ℵ accumulation, too high = loss of degrees of freedom.

## 6. Implications for Clinical Practice

- Therapy as Ψ engineering: CBT/meditation = increase discharge.
- Pharmacology as temporary Ψ boost (but risk dependence).
- Prevention: Lifestyle to maintain Ψ > G (sleep, exercise, low chronic stress).

## 7. Conclusion

The brain does not fail because it is "defective". It fails when neural throughput cannot keep pace with incoming incoherence.

TNA provides a unified, mechanical framework for understanding neural stability—from daily cognition to late-life degeneration.

## Canonical Statement

> *Neural disorders are not mysteries of the mind. They are throughput failures of the brain.*

## The "Plumbing" Axiom of Neuroscience

The brain does not "fail" by chance; it fails due to **throughput saturation**.

> ***The Toilet–Shower Analogy:*** *"Thinking without discharging (sleep/pruning) is the thermodynamic equivalent of a blocked pipe. Eventually, the accumulation ($\aleph$) overflows, destroying the structural integrity ($N_1$) of the house."*